

Skills Summary

Writing Equations from Word Situations

Use this reference while completing your worksheet or when studying.

Situations that Add or Take Away

Rule: give the unknown starting amount a letter, then write what happened to it. “More were added” becomes $x + a$; “some were used, lost or drained” becomes $x - a$. The number after the word *was*, *left* or *now* goes on the other side of the equals sign.

Signal words: more than, added, gained, fewer, left, remaining, fell by.

Example: A jar held j sweets. After 19 were eaten, 27 were left. Find j .

Step 1. The unknown is the starting number and sweets were taken away, so the equation subtracts from j and sets the result equal to what is left: $j - 19 = 27$.

Step 2. Undo the subtraction by adding 19 to both sides: $j = 27 + 19$, so $j = 46$ sweets.

Try it: A shelf held s books. After 23 more were added there were 51. Write the equation and find s .

check: 87

Situations that Share or Scale

Rule: “shared equally between n ” and “cut into n equal pieces” divide the unknown: $\frac{x}{n}$. “ n times as many” and “ n in each of several groups” multiply it: nx . Undo a multiplication by dividing, and a division by multiplying.

Signal words: each, per, shared equally, times as many, equal pieces.

Example: A ribbon r centimetres long is cut into 7 equal pieces of 12 centimetres. Find r .

Step 1. Cutting the whole ribbon into 7 equal pieces divides its length, and one piece is 12, so $\frac{r}{7} = 12$.

Step 2. Undo the division by multiplying both sides by 7: $r = 12 \times 7$, so $r = 84$ centimetres.

Try it: Five identical crates weigh 65 kilograms altogether. Write an equation for the mass m of one crate and find it.

check: 13

Two-Step Situations: a Rate plus a Fixed Charge

Rule: when something is charged *once* and something else repeats, the equation has two terms: (repeating rate) $\times$ (unknown count) + (one time amount) = total, or $ax + b = c$. The one-time amount is never multiplied by the count — that is the mistake this shape is built to catch.

Example: A gym charges 20 dollars to join plus 6 dollars a week. A member has paid 62 dollars. How many weeks w ?

Step 1. The 6 dollars repeats every week so it multiplies w ; the 20 dollars is paid once so it is added on: $6w + 20 = 62$.

Step 2. Subtract 20 from both sides to get $6w = 42$, then divide by 6: $w = 7$ weeks.

Try it: A printer costs 9 dollars to set up plus 4 dollars for each poster. The bill was 41 dollars. Write an equation for the number n of posters and find it.

8 :peep

(!) Watch out: check the equation against the story before you solve it. Substituting your answer back in should make both sides equal — if it does not, the equation is wrong, not the arithmetic.